Equation Sheet: Circuits

Intro Physics II • Lectures 14–17 • Knight ch. 26–28

fundamental = know cold (your starting points); **derived** = a line or two from the fundamentals.

Symbols and constants

I	current (A = C/s)	$\mathcal{E}$	emf (V)	P	power (W)
$V, \Delta V$	potential difference (V)	r	internal resistance (Ω)	Q	charge (C)
R	resistance (Ω)	C	capacitance (F)	τ	time constant (s)
e	1.60×10^{-19} C				

1. Current, resistance & Ohm's law (Knight 26.1–26.5, 27.1–27.4)

fundamental

Current — rate of charge flow $I = \frac{\Delta Q}{\Delta t}$

Ohm's law $I = \frac{\Delta V}{R}$

- for an ohmic resistor; also $\Delta V = IR$ across any R

Resistor power $P = I\,\Delta V$

- rate electrical energy → heat/light

derived

Resistor power, alternate forms $P = I^2R = \dfrac{(\Delta V)^2}{R}$

2. EMF & real batteries (Knight 28.2–28.3)

fundamental

EMF — energy supplied per unit charge $\mathcal{E} = \dfrac{W}{q}$ (units of volts)

derived

Terminal voltage of a real battery $\Delta V_{\text{term}} = \mathcal{E} - Ir$

- internal resistance r; equals $\mathcal{E}$ only when $I = 0$
- real batteries behave like ideal batteries with a single resistor in series

Current from a single-loop source $I = \dfrac{\mathcal{E}}{R + r}$

3. Series & parallel combinations (Knight 28.4)

fundamental

What's shared in series	same I through every element
What's shared in parallel	same ΔV across every element
Resistors in series • voltages add; R_s is bigger than any one	$R_s = R_1 + R_2 + \cdots$
Resistors in parallel • currents add; R_p is smaller than any one	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$
Capacitors • combine the opposite way to resistors	$C_p = C_1 + C_2 + \cdots, \qquad \frac{1}{C_s} = \sum \frac{1}{C_i}$

derived

Two resistors in series	$R_s = R_1 + R_2$
Two resistors in parallel	$R_p = \frac{R_1 R_2}{R_1 + R_2}$
Two capacitors in series	$C_s = \frac{C_1 C_2}{C_1 + C_2}$
Two capacitors in parallel	$C_p = C_1 + C_2$

4. Kirchhoff's rules (Knight 28.1, 28.5)

fundamental

Junction rule (charge conservation)	$\sum I_{in} = \sum I_{out}$
Loop rule (energy conservation) • + across a source from − to +; $-IR$ across a resistor in the direction of I	$\sum_{loop} \Delta V = 0$

5. Meters (Knight 28.5, 28.7)

fundamental

Ammeter • in-line, replacing a wire	ideal: $R = 0$ (in series)
Voltmeter • across the element being measured	ideal: $R = \infty$ (in parallel)

6. RC circuits (Knight 28.6)

fundamental

Capacitor relation	$Q = C\,\Delta V_C$
Time constant	$\tau = RC$

derived

Charging through a battery ΔV_B • starts at $Q = 0$; $I_0 = \Delta V_B/R$ • $Q, \Delta V_C$ climb to 63% of final after one τ; $I, \Delta V_R$ decay from their start	$Q(t) = C\,\Delta V_B\left(1 - e^{-t/\tau}\right), \quad I(t) = I_0\,e^{-t/\tau}$ $\Delta V_R(t) = \Delta V_B\,e^{-t/\tau}, \quad \Delta V_C(t) = \Delta V_B\left(1 - e^{-t/\tau}\right)$
Discharging • starts at Q_0, $\Delta V_0 = Q_0/C$, $I_0 = \Delta V_0/R$ • all four fall to 37% after one τ	$Q(t) = Q_0\,e^{-t/\tau}, \quad I(t) = I_0\,e^{-t/\tau}$ $\Delta V_R(t) = \Delta V_0\,e^{-t/\tau}, \quad \Delta V_C(t) = \Delta V_0\,e^{-t/\tau}$

Companion simulations at `hoppese.github.io/Intro-physics-sims/`: `circuit-builder`, `circuit-lab`, `rc-charging`.